Gauge pinning and a rational boundary Schur gap on a one-sided four-dimensional cell

Project Stardust

October 29, 2026

Abstract

We reconstruct a finite plaquette-incidence problem with 48 vertices, 112 edges and 72 faces. The 32 interior edges have a one-dimensional gauge kernel; pinning one edge gives an invertible 31-coordinate interior Gram matrix without changing its curvature range. Eliminating those interior variables produces a boundary Schur form whose kernel is exactly the 46-dimensional boundary-gradient space. On its 34-dimensional orthogonal complement we prove a computer-assisted rational lower bound 0.69639 and give an exact nonzero Rayleigh witness strictly below 0.6964. Complete geometric and linear-algebra arguments identify the physical norm, cycle compression and residual certificate. The finite arithmetic is supplied as explicit integer tables with an independent standard-library checker. A periodic gauge counterexample and a perturbation cross-term test delimit which invertibility and stability conclusions transfer. This is a finite quadratic construction, not a nonlinear gauge, multiscale or continuum mass-gap theorem; the general Schur and discrete exactness mechanisms are classical.

1 The actual finite problem

Let e_0, e_1, e_2, e_3 be the coordinate unit vectors in $\mathbb{R}^4$. Coordinates and directions in this paper are zero based, matching the formal source. The base vertices are $V_0 = \{0,1\}^4$. The interior oriented edges are

$$E_I = \{(v,i) : v \in V_0,\ v_i = 0\},$$

with source v and target $v + e_i$. There are $4 \cdot 2^3 = 32$ such edges. For each $v \in V_0$ with $v_j = 1$, add a cap vertex $v + e_j$ and an outward edge from v to $v + e_j$. These cap vertices are distinct because they have exactly one coordinate equal to two. There are 32 caps and outward edges.

For every interior edge (v,i) and every $j \neq i$ with $v_j = 1$, add a tangential cap edge from $v + e_j$ to $v + e_i + e_j$. These give 48 tangential edges: choose the ordered distinct directions (i,j) in 12 ways and the remaining two binary coordinates in four ways. Let E_B consist of the 32 outward and 48 tangential edges. There are 80 boundary edges, 48 total vertices V, and 112 total edges $E = E_I \sqcup E_B$.

There are two sorts of faces. The 24 inner squares have base $v \in V_0$, directions $i < j$ and $v_i = v_j = 0$. The 48 ribbons correspond to the triples (v,i,j) used for tangential edges. A ribbon traverses the interior edge (v,i), the outward edge at $v + e_i$ in direction j, the negative tangential cap edge and the negative outward edge at v in direction j. Thus each is a genuine oriented square. Put $F = F_I \sqcup F_R$, of size 72.

Object	Construction	Count
Base/cap vertices	$16 + 32$	48
Interior edges	$4 \cdot 8$	32
Outward/tangential boundary edges	$32 + 48$	80
Inner squares/ribbons	$24 + 48$	72

Source correspondence: `OneSidedBlockIncidence`
`edges_card; innerFaces_card; upperFaces_card`

Source correspondence: `OneSidedCapComplex`
`boundary_card; fullVertex_card; fullEdge_card; fullFace_card`

All edge and face spaces have their ordinary Euclidean norm, with every edge and face counted once. For a potential $\phi \in \mathbb{R}^V$, define $d_0\phi(e) = \phi(\text{target}\, e) - \phi(\text{source}\, e)$. For $f \in \mathbb{R}^E$, let $d_1 f \in \mathbb{R}^F$ be the signed sum around each oriented face. Telescoping around a square gives $d_1 d_0 = 0$. The energy is exactly $\|d_1 f\|^2$, with no weighting or gauge penalty implicitly added.

The source also identifies this cap description with the following direct plaquette rule in $\{0, 1, 2\}^4$: retain all unit squares incident to at least one edge in E_I, and retain their edges. It obtains the same 72 faces and 112 edges, up to row orientation. Reversing a row changes its sign and not the energy. The accompanying checker independently reconstructs this rule and verifies every incidence entry in the certificate.

Source correspondence: `OneSidedCapComplex`
`embedFace_bijective; d1_d0`

Source correspondence: `OneSidedSourceRows`
`row_transport; energy_transport`

2 Exactness by paths, not an assumed gauge quotient

Lemma 2.1 (Every closed field is a gradient)**.** *For this cell, $d_1 f = 0$ if and only if $f = d_0\phi$ for some potential ϕ. The kernel of d_0 consists exactly of constant potentials. Consequently* $\operatorname{rank} d_0 = 47$ *and* $\operatorname{rank} d_1 = 65$.

Proof. For a closed field, first restrict it to the base cube. Define $\phi(v)$ by summing its edge values from the origin to v, flipping the nonzero coordinates in increasing order. Any two orders of these coordinate flips can be related by adjacent interchanges. Interchanging flips in distinct directions changes the path sum by the signed curl around their inner square, which is zero. Thus the sum is independent of their order.

For any base edge (v, i) with $v_i = 0$, a path to $v + e_i$ may therefore be chosen as a path to v followed by that edge. It follows that $\phi(v + e_i) - \phi(v) = f(v, i)$. This proves the gradient identity on every interior edge. For a cap vertex $v + e_j$, define $\phi(v + e_j) = \phi(v) + f(v, j)$, where this last edge is the outward one. This makes the gradient agree with f on all outward edges. The zero-curl equation on each ribbon then forces agreement on its tangential edge as well. These are all edges, so $f = d_0\phi$. The converse is the already established square telescoping identity.

The base cube is connected by coordinate paths, and each cap vertex is joined to its base vertex by an outward edge. Hence the full graph is connected. If $d_0\phi = 0$, potential differences vanish along every edge, so all potential values are the same. Constants conversely have zero gradient. Rank–nullity gives $\operatorname{rank} d_0 = 48 - 1 = 47$. Exactness makes $\dim \ker d_1 = 47$, so $\operatorname{rank} d_1 = 112 - 47 = 65$. □

Source correspondence: `CubePathPoincare`
`path_stokes; cube_poincare`

Source correspondence: `OneSidedCapPoincare`
`d0_recover; closed_iff_gradient`

Source correspondence: `OneSidedGradientRanks`
`gradient_kernel_constant; gradient_rank; curl_rank`

The formal proof implements an ordered-path potential and its exact Stokes error formula. The adjacent-swap proof above reconstructs that same finite path mechanism. No cohomological exactness is assumed merely from the shape of a diagram.

3 The interior gauge and a legitimate pin

Write $A : \mathbb{R}^{E_I} \to \mathbb{R}^F$ for the restriction of d_1 to fields that vanish on the boundary, and $B : \mathbb{R}^{E_B} \to \mathbb{R}^F$ for its boundary restriction. Thus the full curvature is $Af + B\eta$. The interior energy is

$$\|Af\|^2 = \sum_{\text{inner squares}} (\operatorname{curl} f)^2 + \sum_{(v,i,j)\ \text{ribbons}} f(v,i)^2. \tag{1}$$

The second sum occurs because only the interior edge of a ribbon can contribute when the boundary field is zero.

Let $g \in \mathbb{R}^{E_I}$ equal one on the four edges leaving the origin and zero on the other interior edges. Its squared norm is four. It is a full gradient: take the potential ψ equal to -1 at the base origin and zero at every other vertex. None of the boundary edges is incident to the origin, so $d_0\psi$ equals g on the interior and zero on the boundary. In particular $Ag = 0$.

Lemma 3.1 (Exact interior kernel)**.** *The kernel of A is precisely $\mathbb{R}g$; hence* $\operatorname{rank} A = 31$.

Proof. If $Af = 0$, every square and ribbon summand in (1) is zero. An interior edge (v, i) away from the origin has $v_i = 0$ and some other coordinate $v_j = 1$. It belongs to a ribbon, which forces $f(v, i) = 0$. Only the four origin edges can remain. The inner square at the origin in directions i, j now has curl $f(0, i) - f(0, j)$ because its other two edges are away from the origin. Thus all four remaining values agree, so $f = cg$. Conversely $Ag = 0$, proving the kernel statement. Rank–nullity gives $32 - 1 = 31$. □

Source correspondence: `OneSidedBlockKernel`
`kernel_classification; energy_zero_iff`

Source correspondence: `OneSidedBlockRank`
`kernel_span; incidence_rank; gauge_is_gradient`

Choose the pole edge $e_* = (0, 0)$. Pin its value to zero and let $L : \mathbb{R}^{31} \to \mathbb{R}^{72}$ be the resulting interior incidence matrix. For any unpinned f, subtract $f(e_*)g$. This does not change Af and makes the pole value zero. Thus

$$\operatorname{ran} L = \operatorname{ran} A. \tag{2}$$

If a pinned field has zero curvature, Lemma 3.1 says it is cg, and its zero pole value forces $c = 0$. Therefore L is injective and $C = L^T L$ is positive definite. It has an actual inverse: in finite dimension a positive-definite symmetric matrix is injective and hence bijective.

Source correspondence: `OneSidedPinnedInterior`
`lift_pin; pinned_range; pinned_injective; localGram_isUnit`

The unpinned matrix $A^T A$ is singular. Only the proved pin or a separately specified generalized inverse can justify elimination. Equation (2) will allow us to check normal equations in the unpinned coordinates without ever inverting that singular matrix.

4 The boundary gradient and its physical complement

Let $G : \mathbb{R}^{48} \to \mathbb{R}^{80}$ be the restriction of d_0 to boundary edges. Its range is the boundary-gradient space, and define the physical boundary space by

$$\mathcal{P} = (\operatorname{ran} G)^{\perp} = \ker G^T.$$

This is a finite real linear definition. It is not an identification with a nonlinear gauge-invariant quantum state space.

Lemma 4.1 (The complete boundary gauge)**.** *The kernel of G is the two-dimensional space of potentials $\phi = a\mathbf{1} + c\psi$, where ψ is the origin potential defined above. Hence* $\operatorname{rank} G = 46$ *and* $\dim \mathcal{P} = 34$*.*

Proof. Both constants and ψ have zero boundary gradient. They are linearly independent, since ψ is zero at cap vertices and nonzero at the origin. Conversely suppose $G\phi = 0$. Then $d_0\phi$ has only interior components, and its full curl vanishes. Its interior restriction f satisfies $Af = 0$, so $f = cg$ by Lemma 3.1. As full edge fields, $d_0\phi = c\, d_0\psi$. Thus $d_0(\phi - c\psi) = 0$. Lemma 2.1 makes $\phi - c\psi$ constant, proving the complete kernel classification. Rank–nullity gives $48 - 2 = 46$ for G, and ordinary Euclidean orthogonal decomposition gives $\dim \mathcal{P} = 80 - 46 = 34$. □

Source correspondence: `OneSidedBoundaryGauge`
`boundary_kernel; boundary_gradient_rank; boundary_quotient_finrank`

Source correspondence: `OneSidedBoundaryCycles`
`physical_finrank; divergence_zero_iff`

Boundary gauge removal therefore costs 46 dimensions, not the single interior gauge dimension. Subtracting one dimension from both spaces would describe a different boundary problem.

5 Schur elimination and its exact kernel

For a boundary field η, define the pinned response, residual and Schur matrix by

$$\begin{aligned} y_\eta &= -(L^T L)^{-1} L^T B\eta, \\ r_\eta &= L y_\eta + B\eta, \\ S &= B^T B - B^T L (L^T L)^{-1} L^T B. \end{aligned}$$

The normal equation is $L^T r_\eta = 0$. In particular r_η is orthogonal to every pinned interior curvature.

Theorem 5.1 (Completed square and physical kernel)**.** *For all $y \in \mathbb{R}^{31}$ and $\eta \in \mathbb{R}^{80}$,*

$$\|Ly + B\eta\|^2 = \eta^T S \eta + \|L(y - y_\eta)\|^2, \qquad \eta^T S \eta = \|r_\eta\|^2. \tag{3}$$

The minimizing pinned y is uniquely y_η, S is symmetric positive semidefinite, and $\ker S = \operatorname{ran} G$*. Thus* $\operatorname{rank} S = 34$*, and its quadratic form is strictly positive on $\mathcal{P} \setminus \{0\}$.*

Proof. Write $Ly + B\eta = r_\eta + L(y - y_\eta)$. The cross term is zero by the normal equation. Also $B^T r_\eta = S\eta$, and

$$\eta^T S \eta = \langle B\eta, r_\eta \rangle = \langle r_\eta - L y_\eta, r_\eta \rangle = \|r_\eta\|^2.$$

This proves (3). Injectivity of L gives uniqueness of its minimizer. The formula for S is symmetric because $L^T L$ and its inverse are symmetric, and the squared-residual identity proves nonnegativity.

If $\eta \in \operatorname{ran} G$, choose ϕ with $G\phi = \eta$. The full gradient $d_0\phi$ has zero curvature, so some interior completion has $Af + B\eta = 0$. Replace f by its pinned representative using (2). The minimum energy is zero, hence $r_\eta = 0$ and $S\eta = B^T r_\eta = 0$. Conversely, if $S\eta = 0$, then (3) gives $r_\eta = 0$. The full field comprising y_η and η is curl free and therefore a gradient by Lemma 2.1. Its boundary restriction is η, so $\eta \in \operatorname{ran} G$. This proves the kernel equality and the rank. If a physical η has zero energy, its residual is zero and it lies in $\operatorname{ran} G$ as well as its orthogonal complement; testing it against itself gives $\eta = 0$. This proves strict physical positivity. □

Source correspondence: `OneSidedSchurForm`
`completed_square; unique_minimizer; schur_posSemidef`

Source correspondence: `OneSidedSchurForm`
`kernel_schur; schur_rank; physical_positive`

These steps are classical Schur-complement reasoning, applied after a specific gauge pin has been proved legitimate. Gallier's primary exposition [1], Proposition 2.1 and Theorem 4.3, supplies the general positive-definite and singular-block context. Here the kernel and range conditions have been established for the actual incidence problem.

6 Physical cycles and the norm that compression retains

The certificate supplies an integer matrix $C_b \in \mathbb{Z}^{80\times 34}$. Its columns are actual boundary cycles. Two finite identities are crucial:

$$G^T C_b = 0, \qquad (C_b)_{c_i,j} = \mathbf{1}_{i=j} \quad (0 \le i, j < 34), \tag{4}$$

where $c_0, \dots, c_{33}$ are distinct chord-row indices, also supplied. The first identity puts every column in $\mathcal{P}$. The identity minor makes the columns linearly independent: if $C_b z = 0$, reading its chord rows gives $z = 0$. Since $\dim \mathcal{P} = 34$, these columns span all of $\mathcal{P}$. Every physical field is uniquely $\eta = C_b z$.

Source correspondence: `OneSidedBoundaryCycleData`
`cycle_balance; chord_identity`

Source correspondence: `OneSidedBoundaryCycles`
`cycle_injective; cycle_range; physical_representation`

Put

$$H = C_b^T C_b, \qquad K = C_b^T S C_b.$$

Then

$$\|\eta\|^2 = z^T H z, \qquad \eta^T S \eta = z^T K z. \tag{5}$$

In particular H is positive definite. The cycle coordinates are not orthonormal; replacing H by the identity would change the Rayleigh quotient and the claimed physical gap. We keep the generalized quotient $z^T K z/(z^T H z)$ throughout.

Source correspondence: `OneSidedSchurCompression`
`metric_quadratic; compressed_quadratic; gap_transfer`

The orthogonal split into gradient and divergence-free edge fields is classical graph Hodge theory. Jiang–Lim–Yao–Ye [2] describe this mechanism in their primary treatment of edge flows, Theorems 4.7–4.8. Their application is different; no statistical ranking conclusion or universal graph gap is imported into the present plaquette certificate.

7 A normal-equation certificate without a singular inverse

The companion data file contains integer matrices

$$X \in \mathbb{Z}^{32\times 34}, \quad R \in \mathbb{Z}^{72\times 34}, \quad Q \in \mathbb{Z}^{34\times 34}, \quad H \in \mathbb{Z}^{34\times 34}, \quad D = 193028.$$

Here A, B, C_b use the explicit source plaquette ordering in that file. The checked identities are

$$AX + DBC_b = R, \qquad A^T R = 0, \tag{6}$$

$$C_b^T C_b = H, \qquad R^T R = DQ. \tag{7}$$

All are integer equalities. Their exact arithmetic verification is described in Section 9. We first prove their mathematical implication.

Lemma 7.1 (Identifying the certificate with the original Schur form)**.** *Identities* (6)–(7) *imply* $K = Q/D$.

Proof. For any $z \in \mathbb{R}^{34}$, let $\eta = C_b z$ and $f = Xz/D$. The first equation says $Af + B\eta = Rz/D$. The second says this residual is orthogonal to $\operatorname{ran} A = \operatorname{ran} L$. The original minimizing residual r_η also lies in $B\eta + \operatorname{ran} A$ and is orthogonal to that range. The difference of these two residuals lies both in $\operatorname{ran} A$ and its orthogonal complement, so is zero. Therefore $r_\eta = Rz/D$. Taking its squared norm and using (7) gives

$$z^T K z = \|r_\eta\|^2 = \frac{z^T R^T R z}{D^2} = \frac{z^T Q z}{D}.$$

Both matrices are symmetric; equality of their quadratic forms for all z implies equality of matrices, by testing coordinate vectors and their pairwise sums. Hence $K = Q/D$. □

Source correspondence: `OneSidedIntegerResponseData response_equation; normal_equation`

Source correspondence: `OneSidedCertifiedResidual normal; compressed_eq_residual_gram; compressed_entry`

Source correspondence: `OneSidedBoundaryGap compressed_as_data; metric_as_data`

Notice that X uses all 32 unpinned coordinates. Its normal equations are valid without an inverse of $A^T A$. Uniqueness of the *residual*, not uniqueness of the unpinned response, is what identifies the certificate. Row orientation signs in the native cap model cancel in the residual Gram and normal equations; the checker independently verifies the canonical lattice row geometry.

8 The rational lower certificate and a sub-rounded witness

Define the integer matrix

$$T = 25000Q - 69639 \cdot 48257H. \tag{8}$$

The certificate supplies a positive integer s, positive integers $w_0, \ldots, w_{33}$ and an integer matrix $V \in \mathbb{Z}^{34\times 34}$ such that

$$sT = V \operatorname{diag}(w_0, \ldots, w_{33}) V^T. \tag{9}$$

The exact value of s has 2,386 decimal digits. All of it, every weight and every matrix entry are provided in the accompanying data; no floating-point factorization or approximate positivity check is used.

Theorem 8.1 (Certified physical boundary gap)**.** *For every $\eta \in \mathcal{P}$,*

$$\frac{69639}{100000}\|\eta\|^2 \le \eta^T S \eta.$$

If λ_* is the minimum physical Rayleigh quotient, then

$$\frac{69639}{100000} \le \lambda_* \le \frac{729067245}{1046910149} < \frac{1741}{2500} = 0.6964. \tag{10}$$

Proof. For any z, (9) gives $sz^T Tz = \sum_k w_k((V^T z)_k)^2 \ge 0$. Division by $s > 0$ gives $z^T Tz \ge 0$. Since $D = 4 \cdot 48257$, direct substitution into (8) yields

$$K - \frac{69639}{100000}H = \frac{T}{25000D} = \frac{T}{4825700000}.$$

The denominator is positive. Equations (5) therefore prove the gap for every physical field.

For the upper comparison, the certificate's explicit vector is

$$\begin{aligned} z_* = (&2000, 0, 2000, 760, 2000, 2000, -704, -704, 9, -1935,\\ &-704, -704, -1935, 9, -1935, 9, -130, -130, -1935, -704,\\ &-704, -1935, -1935, 0, 0, -56, -56, -56, -56, -130,\\ &-56, -56, -130, -130)^T. \end{aligned}$$

The exact integer dot products are

$$z_*^T H z_* = 117410484, \qquad z_*^T Q z_* = 15782847719760.$$

Thus $\eta_* = C_b z_*$ is physical and nonzero, and its quotient is

$$\frac{\eta_*^T S \eta_*}{\|\eta_*\|^2} = \frac{15782847719760}{193028 \cdot 117410484} = \frac{729067245}{1046910149} < \frac{1741}{2500}.$$

The final comparison is exact cross multiplication of positive integers. The minimum exists: the unit sphere of the finite-dimensional physical space is compact and the quadratic form is continuous. Its lower and upper bounds follow from the proved inequality and this test field. □

Source correspondence: `OneSidedBoundaryPositiveData`
`scale_positive; weights_positive; weighted_gram_identity`

Source correspondence: `OneSidedBoundaryGap`
`corrected_eq; compressed_bound; boundary_gap`

Source correspondence: `OneSidedBoundaryGapWitness`
`integer_norm; integer_gram; rayleigh_value; below_rounded`

The witness quotient is approximately 0.69639906127; this decimal is only an orientation aid. In particular the rounded decimal 0.6964 is *not* a valid lower bound, even though the certified interval is less than 10^{-5} wide. The rational lower and upper tests have different roles. No assertion that the witness is an exact eigenvector or minimizer is made. The compactness statement introducing λ_* is a classical human rephrasing of the mapped quadratic-form bounds, not a new formal declaration.

9 What the finite arithmetic checks

This theorem is computer assisted at its finite integer-certificate stage. The geometric exactness, gauge pin, completed square, norm identification and deduction from a positive Gram certificate were proved above. The remaining finite input is fully specified by `certificate-data.json` and checked by `verify_certificate.py`. The JSON contains ordinary integer arrays and their original source hashes. Its SHA-256 digest, using UTF-8 text with LF newlines, is

`b1a7d776078753d95a18c451f4536b6cc4aeea2b12142fedaf8e66b972d01078`.

It includes the interior and boundary edge coordinates, source face coordinates, boundary endpoint indices, chord indices, A, B, C_b, X, R, H, Q, T, s, w, V and z_*. Directions and array indices are zero based.

The checker uses only Python's standard library and unbounded integer arithmetic, together with exact rational arithmetic for the last quotient. Its logical operations are the following finite computations.

1. Generate all unit squares in $\{0,1,2\}^4$ incident to one of the 32 interior edges; verify the 72-face, 112-edge, 48-vertex geometry and the 32/80 split against the listed coordinate arrays.
2. For every face, assign the four coefficients $+1, +1, -1, -1$ along its canonical boundary. Verify every entry of A, B against those rows.
3. Construct the boundary gradient matrix from its actual source and target vertices. Verify both equations in (4), including all 34 chord rows.
4. Multiply the integer arrays to verify every entry of (6) and (7).
5. Verify (8), check $s > 0$ and every $w_k > 0$, and verify all 1,156 entries of (9) by ordinary integer multiplication and addition.
6. Verify the two witness dot products and the rational comparisons in (10) by exact integer or fraction arithmetic.

Each matrix product uses the defining formula $(MN)_{ij} = \sum_k M_{ik}N_{kj}$. Thus no eigenvalue solver, numerical tolerance, symbolic positivity oracle or unproved matrix inversion is hidden in the finite certificate. These checks pass independently of the source's kernel-checked integer identities. The companion receipt records the data hash and the actual successful check; it is not a substitute for the analytic and geometric implications of those identities.

10 A local pin is not global unpinned invertibility

There is a concrete periodic counterexample to transferring a local pinned statement to simultaneous elimination of all unpinned block interiors. Let the periodic vertices be $(\mathbb{Z}/4\mathbb{Z})^4$, with positive coordinate edges. Partition them into side-two blocks by the coordinate labels $\lfloor v_i/2 \rfloor$. Call an edge interior if its two endpoints have the same four block labels. Define a periodic potential

$$\Phi(v) = \begin{cases} 1, & v_0 \in \{1,2\}, \\ 0, & v_0 \in \{0,3\}. \end{cases}$$

Let $f_i(v) = \Phi(v + e_i) - \Phi(v)$. It vanishes in directions $i \neq 0$. In direction zero, the values for $v_0 = 0, 1, 2, 3$ are respectively $1, 0, -1, 0$. The nonzero edges run from zero to one and from two to

three, so they are interior edges. The crossing edges from one to two and from three to zero have zero value. Hence f is nonzero and entirely supported on the simultaneously retained interiors.

Every plaquette curl of f is zero, because it is a gradient and the coordinate shifts commute. If A_{per} denotes the full periodic plaquette incidence restricted to all these interior edges, its nonzero interior field f_I satisfies $A_{\mathrm{per}} f_I = 0$. Consequently $A_{\mathrm{per}}^T A_{\mathrm{per}} f_I = 0$. It has no left inverse and no strictly positive unpinned coercivity constant: either assertion applied to f_I would force it to vanish or give positive norm bounded by zero.

Source correspondence: `PeriodicInteriorGaugeKernel`
`gauge_boundary_zero; witness_nonzero; incidence_witness_zero`

Source correspondence: `PeriodicInteriorGaugeKernel`
`hessian_witness_zero; hessian_no_left_inverse`

This is compatible with the local result. The one-sided pinned problem and the unrestricted periodic all-block interior matrix are different operators. The counterexample does not exclude a carefully defined global gauge quotient or a different pinned global estimate; those would require their own domain, kernel and coercivity proofs.

11 A separate perturbation boundary: missing linear loss

Even finite positive blocks do not allow a general perturbation estimate that drops coupling cross terms. For a scalar block

$$H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad c > 0, \qquad \mathcal{S}(a,b,c) = a - b^2/c,$$

changing b to $b + d$ at $c = 1$ gives the exact identity

$$\mathcal{S}(a, b+d, 1) = \mathcal{S}(a,b,1) - 2bd - d^2. \tag{11}$$

At $a = 2, b = 1, d = 1/10$, both blocks are positive definite: their Schur values are 1 and $79/100$. But

$$\frac{79}{100} < 1 - 4(1/10)^2 = \frac{96}{100}.$$

Thus a proposed loss of only $4d^2$ relative to an arbitrary nonzero reference coupling is already false in dimension two. More generally $2d + d^2 > 4d^2$ for $0 < d < 2/3$, so arbitrarily small positive perturbations have the same defect. For $|d| \le \varepsilon$, (11) instead gives the valid bound

$$\mathcal{S}(a, b+d, 1) \ge \mathcal{S}(a,b,1) - 2|b|\varepsilon - \varepsilon^2.$$

Source correspondence: `ManuscriptSchurDiagnostics`
`fixed_interior_increment; general_stability_counterexample`

Source correspondence: `ManuscriptSchurDiagnostics`
`small_coupling_counterexample; fixed_interior_lower_bound`

For comparison, quadratic loss is justified at a zero reference coupling. If a real Hilbert-space block energy is

$$\mathcal{E}(x,y) = ax^2 + 2x\langle b, y\rangle + \langle Cy, y\rangle,$$

with $a \ge a_0 - \delta$, $\|b\| \le \varepsilon$ and $C \ge mI$, $m > 0$, then completing the scalar square yields

$$\begin{aligned} \mathcal{E}(x,y) &\ge (a_0 - \delta)x^2 - 2|x|\varepsilon\|y\| + m\|y\|^2 \\ &\ge \left(a_0 - \delta - \frac{\varepsilon^2}{m}\right) x^2. \end{aligned}$$

If instead $C_0 \geq m_0 I$ and $\|C - C_0\| \leq \eta \leq m_0/2$, Cauchy–Schwarz gives $\langle Cy, y\rangle \geq (m_0 - \eta)\|y\|^2 \geq (m_0/2)\|y\|^2$. The same proof gives loss $\delta + 2\varepsilon^2/m_0$. These are conditional perturbation statements, with a zero-coupling reference and proved interior coercivity. They do not automatically transport the numerical cell gap to nonlinear fields or other geometries.

Source correspondence: `CoerciveSchurStability`
`operator_quadratic_perturbation; operator_robust_lower`

12 Provenance, significance and remaining uniformity questions

The mathematics is reconstructed from frozen verified commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The exact declaration map and original source hashes accompany the paper. The independent checker verifies a literal extraction of the frozen integer tables against geometry; no mathematical research source, configuration or original manuscript was altered. The source's complete build and axiom audit receipt was checked by its evidence hashes; no new research build is claimed.

Finite exactness, gauge gradients, Schur complements and positive weighted Gram certificates use classical mechanisms. The primary comparisons above establish that context, not novelty of general methods. Priority for the specific one-sided geometry and exact narrow bracket remains unresolved. The result is a reproducible geometry-specific finite quadratic estimate, including an explicit correction to rounding and a sharp distinction between local pinning and a global unpinned matrix.

There is no varying block size, lattice spacing, volume or continuum limit in the theorem. It neither constructs a four-dimensional quantum field nor identifies a physical correlation or Hamiltonian spectrum. The linear boundary complement is not silently substituted for a nonlinear gauge quotient. Extending the estimate would require exact compatible global gauge conditions, control of interactions and approximation errors, and estimates uniform in the particular scale and volume parameters. The proved finite gap discharges none of these separate analytic obligations.

References

[1] J. Gallier, *The Schur complement and symmetric positive semidefinite (and definite) matrices*, author notes, August 24, 2019. Primary author PDF.

[2] X. Jiang, L.-H. Lim, Y. Yao and Y. Ye, *Statistical ranking and combinatorial Hodge theory*, primary preprint, arXiv:0811.1067.